

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination - 2023

Course: Second Year B. Tech. (Sem III)

Branch: Civil Engineering

Subject Name: Hydraulics-I

Subject Code: BTCVC304

Max Marks: 60

Duration: 3 Hrs.

Date: 09/01/2024

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) (1) Explain the phenomenon of the capillarity. Obtain the expression of capillary rise of a liquid.

CO2

6

(2) At a certain point in castor oil the shear stress is 0.432 N/m^2 and the velocity gradient 0.216 s^{-1} . If the mass density of castor oil is 959.42 kg/m^3 , find kinematic viscosity.

- B) An inverted U-tube differential manometer is connected to two horizontal pipes A and B through which water is flowing. The vertical distance between the axes of these pipes is 30 cm. When an oil of specific gravity 0.8 is used as a gauge fluid, the vertical heights of water columns in the two limbs of the inverted manometer (when measured from the respective centre of the pipe) are found to be same i.e. 35cm. Determine the pressure between the pipe.

CO2

6

- C) Define buoyancy and centre of buoyancy. Derive an expression for the metacentric height of a floating body.

CO2

6

Q.2 Solve Any Two of the following.

12

- A) A 30 cm diameter pipe, conveying water, branches into two pipes of diameters 20 cm and 15 cm respectively. If the average velocity in 30 cm diameter pipe is 2.5 m/s, find the discharge in this pipe. Also determine the velocity in 15 cm pipe if the average velocity in 20 cm diameter pipe is 2 m/s.

CO1

6

- B) The velocity potential function is given by an expression of

CO1

6

$$\phi = -\frac{xy^3}{3} - x^2 + \frac{x^3y}{3} + y^2$$

- a. Find the velocity components in x and y directions respectively.
- b. Show that ϕ represents a possible case of flow.

- C) Derive Euler's equation of motion. Explain how this is integrated to get Bernoulli's equation.

CO1

6

Q. 3 Solve Any Two of the following.		12
A) Derive the expression for shear stress distribution and velocity distribution in fully developed laminar flow through pipe.	CO4	6
B) Write a short note on Boundary Layer Theory.	CO4	6
C) What is Prandtl mixing length theory? Find an expression for shear stress due to Prandtl.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain concept of Rayleigh's method and find an expression for the time period using Rayleigh's method, where time period (t) of a pendulum depends upon the length (L) of the pendulum and acceleration due to gravity (g).	CO4	6
B) Give short explanation about following non-dimensional numbers – (i) Froude number, (ii) Euler Number, (iii) Reynold's Number, (iv) Mach number, (v) Weber number	CO4	6
C) Compare between distorted and undistorted models with the suitable example.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) What is Compound Pipe? What will be loss of head when pipes are connected in series?	CO3	6
B) Describe Water Hammer and write a note on Surge Tank.	CO3	6
C) What are the uses of Nomogram and Moody's chart?	CO3	6

***** End *****